

Project Title: Autonomous In-Campus Robo-Delivery System using ESP32 and Standalone Web Interface

Abstract:

This project proposes an autonomous robotic delivery system designed specifically for the National Institute of Technology Goa (NITG) campus. Traditional delivery methods within closed college campuses often face logistical delays. To address this, we are developing a cost-effective, self-navigating rover powered by an ESP32 microcontroller. A key feature of this system is its completely standalone network architecture. The ESP32 operates in Access Point (AP) mode, hosting a local web server that allows users to dispatch the robot using a smartphone without requiring an active internet connection.

The user interface is a mobile-optimized web application embedded directly into the flash memory of the ESP32. Users can select predefined campus waypoints, such as Terekhol Hostel or the Admin Block, to initiate a delivery sequence. By utilizing Base64 encoding and direct flash memory streaming, the system efficiently serves a graphical interface entirely offline.

For physical navigation, the robot integrates a U-blox NEO-6M GPS module for coordinate tracking, an I2C magnetometer for directional heading, and an array of HC-SR04 ultrasonic sensors for real-time obstacle avoidance. Currently, the core software architecture, offline web interface, and memory management systems have been successfully developed. The final phase of the project will focus on the hardware integration of the motor drivers and sensor arrays to achieve full autonomous mobility. Ultimately, this project demonstrates a highly efficient, internet-independent solution for localized logistics within academic environments.